

23. human MYB DNA_BIND "H-T-H motif" (24 aa) and arabidopsis Q6R032

TOOLS: kmerseek hp_kyte_doolittle2 k21 (chosen for DNA_BIND in Stage 0) and hp_thomas_dill2 k19; phmmer, MMseqs2, MMseqs2 iterative, Foldseek, ProstT5, Reseek; Kyte-Doolittle scan

- kmerseek: call lands on the feature (>= 80% of call inside, >= 30% covered)

call mostly inside the feature, lower IoU than kmerseek

call overlaps the feature but < 50% of the call is inside it

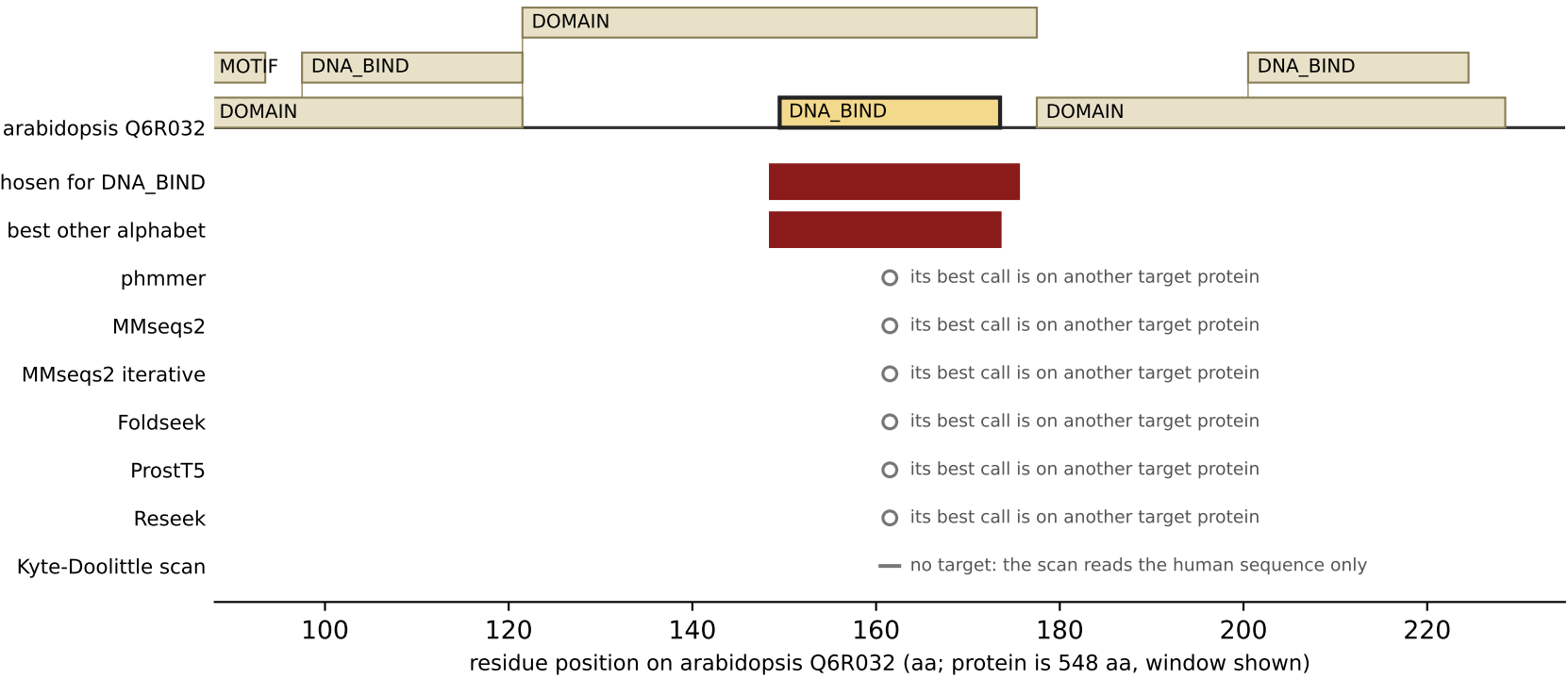
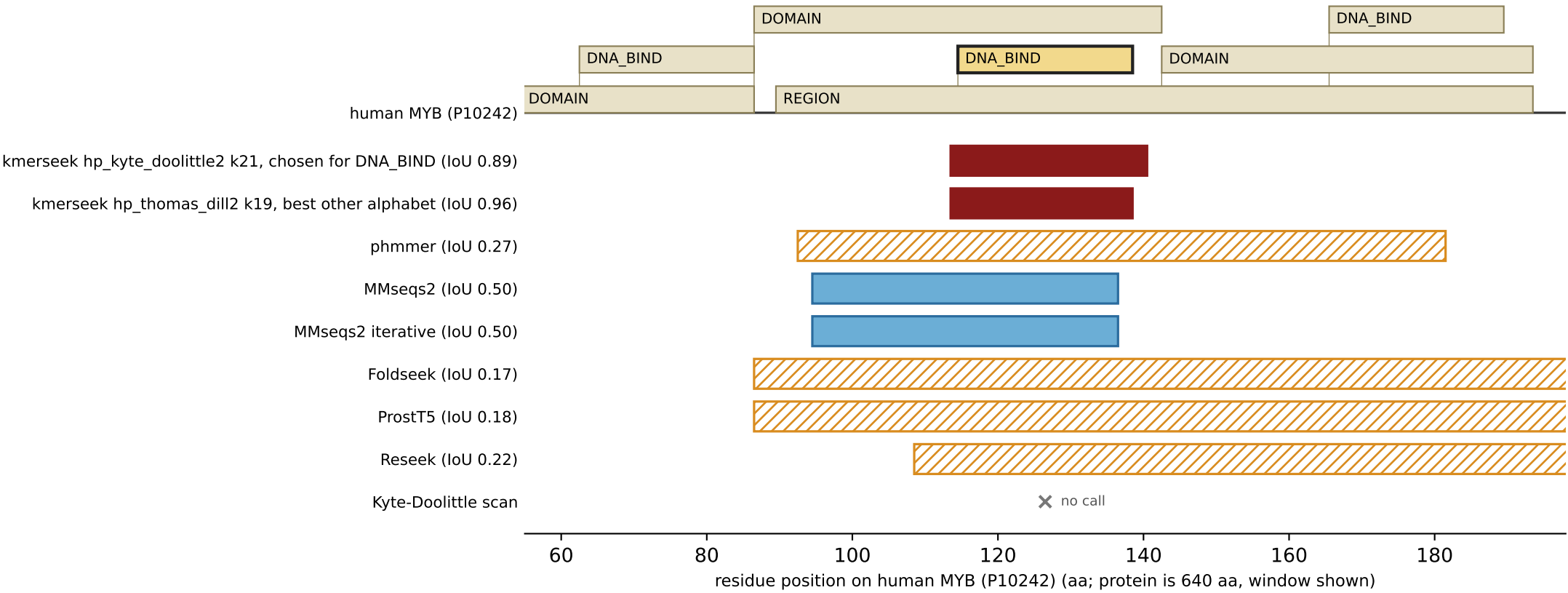
call reaches kmerseek's IoU or higher

no call of this feature type overlaps the feature
- the tool's best call is on another target protein

Kyte-Doolittle scan: no target protein, it searches nothing

the Swiss-Prot feature being scored (human) / transferred from (target)

other Swiss-Prot features, labelled by type



Hypothesis: kmerseek places the human DNA_BIND feature "H-T-H motif" on its arabidopsis counterpart and the structure tools do not. Conclusion: kmerseek IoU 0.89, 24 of 27 residues identical; Foldseek spills; ProstT5 spills; Reseek spills; phmmer spills; MMseqs2 inside, lower IoU; MMseqs2 iterative inside, lower IoU; Kyte-Doolittle scan did not land; pLDDT 96, disorder fraction 0.0.